

Analyzing the Impact of Car Features on Price and Profitability

A Data-Driven Study of Automotive Market Dynamics and Pricing Strategy

Prepared By

****Surendra Kumar ****

Program

MSc Big Data Analytics

Project Type

Data Analytics Portfolio Project

Abstract

The automotive industry is undergoing a period of rapid transformation driven by technological innovation, environmental sustainability, and evolving consumer preferences. Manufacturers must continuously balance product performance, pricing strategies, and market demand to maintain profitability and competitiveness.

This project investigates the relationship between vehicle characteristics, market categories, and pricing using a dataset containing over 11,000 vehicle records. Through exploratory data analysis, statistical techniques, regression analysis, and interactive dashboard development, the study identifies the key factors influencing Manufacturer Suggested Retail Price (MSRP) and profitability.

The findings provide valuable recommendations for optimizing pricing strategies, product development decisions, and market positioning within the automotive sector.

Keywords: Automotive Analytics, Pricing Strategy, Profitability Analysis, Market Segmentation, Business Intelligence, Data Analytics.

1. Introduction

The global automotive market is characterized by intense competition, rapid technological advancements, and changing consumer expectations. The growing adoption of hybrid and electric vehicles, combined with increasing demand for fuel efficiency and advanced vehicle technologies, has transformed the strategic priorities of manufacturers.

To remain competitive, organizations must understand how vehicle specifications, performance characteristics, and market positioning influence pricing and consumer demand. Data analytics provides an effective approach for identifying these relationships and supporting evidence-based business decisions.

This project examines how vehicle features impact pricing and profitability, enabling manufacturers to optimize their product portfolios and pricing strategies.

2. Project Objectives

The primary objectives of this study were:

- To analyze the relationship between vehicle characteristics and pricing.
 - To identify the most influential features affecting MSRP.
 - To investigate vehicle popularity across different market categories.
 - To evaluate the relationship between engine performance and profitability.
 - To examine fuel efficiency trends across vehicle segments.
 - To provide strategic recommendations for product development and pricing optimization.
-

3. Dataset Overview

The analysis was conducted using the **Car Features and MSRP Dataset**.

Dataset Characteristics

Metric	Value
Number of Records	11,159
Number of Variables	16
Industry	Automotive
Data Source	Kaggle (Cooper Union Dataset)

Key Variables

- Make
 - Model
 - Year
 - Engine Fuel Type
 - Engine Horsepower
 - Engine Cylinders
 - Transmission Type
 - Driven Wheels
 - Number of Doors
 - Market Category
 - Vehicle Size
 - Vehicle Style
 - Highway MPG
 - City MPG
 - Popularity
 - MSRP
-

4. Methodology

4.1 Data Preparation

Prior to analysis, the dataset was subjected to data quality checks including:

- Missing value assessment
- Duplicate record identification
- Data validation
- Outlier detection and treatment

These preprocessing steps ensured analytical accuracy and reliability.

4.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution and relationships among variables.

The analysis included:

- Vehicle pricing distribution
- Market category analysis
- Fuel type comparison
- Manufacturer performance comparison

- Feature popularity assessment
-

4.3 Statistical Analysis

Several statistical techniques were applied:

Correlation Analysis

Correlation coefficients were calculated to quantify relationships between vehicle features and pricing.

Regression Analysis

Multiple regression techniques were used to identify variables that significantly influence MSRP.

Market Segmentation

Vehicles were segmented into market categories to evaluate profitability and demand patterns.

4.4 Dashboard Development

An interactive dashboard was created using Microsoft Excel featuring:

- Pivot Tables
- Pivot Charts
- Dynamic Filters
- Slicers
- Interactive KPI Visualizations

The dashboard enables users to explore vehicle performance across multiple dimensions.

5. Analysis and Findings

5.1 Market Category and Popularity

The analysis revealed substantial variation in vehicle popularity across market categories.

Key Finding

Performance-oriented and luxury vehicle segments demonstrated higher popularity scores, while economy-focused vehicles maintained broader market coverage.

Business Implication

Market category strongly influences customer perception and purchasing behaviour.

5.2 Engine Performance and Pricing

A scatter plot analysis examined the relationship between engine horsepower and MSRP.

Key Finding

A positive relationship exists between engine power and vehicle price, indicating that performance remains a significant value driver.

Business Implication

Consumers are willing to pay premium prices for enhanced vehicle performance.

5.3 Feature Importance Analysis

Regression analysis identified the variables with the strongest influence on pricing.

Significant Factors

- Engine Horsepower
- Market Category
- Vehicle Size
- Fuel Type
- Transmission Type

Business Implication

These factors should be prioritized when developing pricing strategies and future vehicle models.

5.4 Manufacturer Pricing Analysis

Average MSRP was evaluated across vehicle manufacturers.

Key Finding

Premium automotive brands consistently achieved higher average pricing than mass-market manufacturers.

Business Implication

Brand positioning significantly contributes to perceived vehicle value.

5.5 Fuel Efficiency Analysis

The relationship between engine cylinders and fuel efficiency was examined.

Key Finding

Vehicles with larger engines generally exhibited lower fuel efficiency compared to smaller-engine alternatives.

Business Implication

Fuel efficiency remains an important consideration for consumers and regulatory compliance.

6. Discussion

The analysis demonstrates that vehicle pricing is influenced by both technical specifications and market positioning.

Higher performance vehicles typically command premium prices; however, profitability depends on balancing production costs with consumer demand. Market segmentation revealed that mid-range SUV categories offer an attractive combination of demand and profitability.

The findings also indicate increasing opportunities within hybrid and electric vehicle segments as consumer preferences continue to evolve toward sustainable transportation solutions.

7. Strategic Recommendations

Based on the analytical findings, the following recommendations are proposed:

1. Focus on High-Profit Market Segments

Expand product offerings within the mid-range SUV and crossover categories.

2. Increase Investment in Alternative Fuel Technologies

Strengthen research and development initiatives for hybrid and electric vehicles.

3. Implement Feature-Based Pricing Strategies

Align pricing with features that deliver the greatest perceived customer value.

4. Improve Fuel Efficiency

Develop powertrain technologies that balance performance with environmental sustainability.

5. Leverage Brand Positioning

Enhance brand value through innovation, quality, and customer experience.

8. Skills Demonstrated

- Data Cleaning
 - Exploratory Data Analysis (EDA)
 - Correlation Analysis
 - Regression Analysis
 - Dashboard Development
 - Data Visualization
 - Market Segmentation
 - Pricing Analytics
 - Business Intelligence
 - Decision Support Analytics
-

9. Conclusion

This project demonstrates the practical application of data analytics within the automotive industry. Through the analysis of over 11,000 vehicle records, valuable insights were generated regarding pricing drivers, vehicle popularity, fuel efficiency, and profitability.

The findings highlight the importance of combining technical innovation with data-driven decision-making to optimize product development and pricing strategies. The project further strengthened analytical, visualization, and business intelligence skills that are highly relevant to modern Data Analyst and Business Intelligence roles.

References

1. Kaggle – Car Features and MSRP Dataset
2. Microsoft Excel Documentation
3. Automotive Industry Market Reports

4. Trainity Data Analytics Program